PRODUCT GUIDE

Lumora Thermostat T2 Product Guide

Document number	LHS-PG-T2-001
Revision	3.1
Effective date	March 2, 2026
Document owner	Product Support Engineering
Audience	Customer Care, Technical Support, Certified Installers
Classification	Internal - do not distribute outside Lumora

Purpose. Reference for supporting the Lumora Thermostat T2 (model LT-T2), covering specifications, installation, configuration, troubleshooting and error codes. Use this guide together with the Ticket Escalation SOP when a case cannot be resolved at first contact.

1. Product Overview

The Lumora Thermostat T2 is the entry model of the Lumora climate line. It replaces a standard low-voltage wall thermostat and adds scheduling, remote control through the Lumora Home app and monthly energy reports. The T2 is designed for single-zone homes with conventional furnaces, boilers and single-stage heat pumps.

The T2 is sold under SKU LT-T2-01 at a list price of $149.00. It is positioned below the Lumora Thermostat T3, which adds a color touchscreen, a humidity sensor, multi-stage heat pump support and battery-backed operation. Agents should take care not to mix up the two models: several limits, error codes and the warranty term are different.

1.1 What Is in the Box

- Thermostat T2 display unit and wall plate
- Optional trim plate for covering marks left by older thermostats
- Two wall screws, two drywall anchors and a sheet of wire labels
- Quick Start card with the pairing code (also printed on the back of the unit)

1.2 System Compatibility

The T2 works with 24 VAC low-voltage systems only. It is compatible with gas, oil and electric furnaces, hot-water and steam boilers, central air conditioning and single-stage heat pumps without auxiliary heat.

The T2 requires a C-wire (24 VAC common) for continuous power. If the home has no C-wire, the Lumora Power Extender Kit PEK-1 must be installed at the equipment control board. The T2 has no internal battery and cannot run on power-stealing.

Warning: The T2 is not compatible with line-voltage (120 V or 240 V) systems such as electric baseboard heaters. Thick wires joined with wire nuts at the old thermostat usually indicate line voltage. Do not proceed with installation and advise the customer that the product is not suitable.

2. Specifications

Item	Thermostat T2 (LT-T2)
Display	2.8 in monochrome LCD with backlight, push-and-turn control ring
Setpoint range	45 °F to 90 °F (7 °C to 32 °C)
Temperature accuracy	±1.0 °F (±0.5 °C)
Humidity sensing	Not available
Supported stages	Up to 2 heat / 1 cool (conventional); 1-stage heat pump without auxiliary heat
Power	24 VAC, 50/60 Hz; C-wire required (or PEK-1 kit)
Power consumption	Max 1.2 W
Wireless	802.11 b/g/n, 2.4 GHz only; LumoraLink radio for sensors
Room Sensor S1 support	Up to 2 sensors
Compressor protection	Fixed 5-minute minimum off-time
Clock backup	Retains time for 48 hours without power
Dimensions	3.9 x 3.9 x 1.1 in (99 x 99 x 28 mm)
Operating environment	32 °F to 104 °F, 5% to 90% RH non-condensing
Warranty	2 years (24 months) limited warranty

Table 2-1. Thermostat T2 technical specifications.

Settings and schedules are stored in non-volatile memory and survive a power loss of any duration. Only the clock depends on the backup capacitor; after more than 48 hours without power the T2 resynchronizes its clock from the network when it reconnects.

3. Installation

Installation takes 30 to 45 minutes for most homes. Customers may install the T2 themselves. Customer Care may offer professional installation through the Lumora Certified Installer network for a flat fee of $129.

3.1 Before You Start

1. Switch off power to the heating and cooling equipment at the circuit breaker or service switch. Confirm the system is off by raising the setpoint on the old thermostat and waiting two minutes; the equipment must not start.
2. Remove the faceplate of the existing thermostat and take a clear photo of the wiring before disconnecting anything. Installers must attach this photo to the installation record.
3. Label each wire with the supplied wire labels according to the terminal letter it was connected to, not by wire color. Wire colors are not standardized between installers.
4. Disconnect the wires and remove the old wall plate. Wrap the wire bundle around a pencil so that it cannot fall back into the wall.

3.2 Wiring Terminals

Terminal	Function	Typical wire color
R / Rh	24 VAC power (heating transformer)	Red
Rc	24 VAC power (cooling transformer); jumper to R if only one	Red
C	24 VAC common - required for the T2	Blue or black
W1	Heat stage 1	White
W2	Heat stage 2	Brown or white
Y1	Cool stage 1 / compressor	Yellow
G	Fan	Green
O/B	Heat pump reversing valve	Orange or dark blue

Table 3-1. T2 wall plate terminals.

3.3 Mounting and Wiring

1. Hold the T2 wall plate against the wall with the bubble level at the top, mark the two screw holes and drill 3/16 in holes if anchors are needed.
2. Feed the wires through the center opening and screw the wall plate to the wall.
3. Insert each labelled wire into the matching terminal until the terminal button clicks. Strip wires to 1/4 in if the copper end is damaged.
4. Push excess wire back into the wall and snap the display onto the wall plate.
5. Restore power at the breaker. The T2 starts within 60 seconds and shows the setup screen.

3.4 Power Extender Kit PEK-1

If the customer has no C-wire, the PEK-1 is installed inside the furnace or air handler and creates a virtual common conductor over the existing wires. The kit is free of charge for T2 customers who request it within 60 days of delivery; after that it costs $24.99. Only Tier 2 Technical Support may authorize a free PEK-1 outside the 60-day window.

4. Setup and Configuration

4.1 First-Time Setup

1. Choose the language and temperature unit on the thermostat.
2. Select the equipment type: conventional (furnace or boiler) or heat pump.
3. Connect the T2 to the home wireless network (2.4 GHz only). Networks that use the same name on 2.4 GHz and 5 GHz work as long as 2.4 GHz is enabled.
4. Open the Lumora Home app, tap Add Device and enter the six-digit pairing code from the Quick Start card.
5. Run the equipment test from Settings > Equipment > Test to confirm heat, cool and fan each start correctly.

4.2 Schedules and Comfort Settings

The T2 supports up to four schedule periods per day (Wake, Away, Home, Sleep) with separate weekday and weekend programs. Temporary holds last until the next schedule period; permanent holds stay in place until the customer cancels them in the app or on the device.

Auto changeover between heating and cooling requires a minimum deadband of 3 °F between the heat and cool setpoints. The T2 refuses settings that would make the deadband smaller.

4.3 Adding Room Sensors

The T2 can average its own reading with up to two Lumora Room Sensor S1 units. When a third sensor is added the app shows the message "Sensor limit reached". Customers who need more sensors should be offered the T3, which supports more sensors.

5. Troubleshooting

Always confirm the firmware version (Settings > About) before troubleshooting. The current release for the T2 is 4.8.2. Devices on 4.6 or earlier must be updated first because several connection issues were fixed in 4.7.

Symptom	Likely cause	Resolution
Display stays blank after installation	No 24 VAC on R and C; furnace door switch open; blown 3 A fuse on control board	Check the furnace door is fully closed. Measure R-C voltage. Replace the fuse if needed. Escalate to Tier 2 if voltage is present.
System short-cycles	Thermostat mounted near a supply vent or in direct sunlight	Relocate or increase the cycle rate setting. Compressor off-time is fixed at 5 minutes and cannot be lowered.
Temperature reads 2-4 °F high	Heat from a lamp, TV or exterior wall	Apply a temperature offset of up to ±5 °F in Settings > Sensors.
Fan runs constantly	G wire touching another terminal or fan set to On	Set fan to Auto; reseat the G wire.
App shows device offline	Router changed or 5 GHz-only network	Reconnect from Settings > Network; see error E4.

Table 5-1. Common T2 symptoms.

5.1 Factory Reset

A factory reset erases schedules, network credentials and the link to the customer account. To reset the T2, press and hold the control ring for 12 seconds until the display shows RESET, then release. The device restarts in setup mode. Agents should only suggest a factory reset after the standard troubleshooting steps fail, and must note it in the ticket.

6. Error Codes

Error codes appear in the top-right corner of the display and in the app. The same code number can mean something different on the T3; always check the model before looking up a code.

Code	Meaning	Recommended action
E1	Internal temperature sensor failure	Power-cycle; if it persists, start a warranty claim.
E2	No heat detected after 20 minutes of heat call	Check furnace operation and the W1 wire.
E3	No cooling detected after 20 minutes of cool call	Check Y1 wire and outdoor unit breaker.
E4	Lost connection to the wireless network for more than 10 minutes	Confirm 2.4 GHz is enabled; reconnect in Settings > Network.
E5	Room sensor not reporting for 2 hours	Replace the S1 battery or move the sensor closer.
E6	Wall plate contact fault	Remove and reseat the display on the wall plate.

Code	Meaning	Recommended action
E7	Low supply voltage: R-C below 18 VAC	Check C-wire and transformer; install PEK-1 if there is no C-wire.
E8	Firmware update failed	Device rolls back automatically; retry from the app after 1 hour.
E9	Clock not set after power loss	Reconnect to the network to sync time.

Table 6-1. Thermostat T2 error codes.

7. Maintenance, Warranty and Support

7.1 Maintenance

The T2 needs no routine maintenance. Clean the display with a dry microfiber cloth; do not spray liquid cleaners on the unit. Firmware updates install automatically overnight when the thermostat is idle.

7.2 Warranty Summary

The Thermostat T2 carries a 2-year (24-month) limited warranty from the date of delivery. This is shorter than the T3 warranty. Full terms, exclusions and the claim procedure are defined in the Lumora Limited Warranty Policy (LHS-POL-001) and the Warranty Claims SOP (LHS-SOP-TS-007).

7.3 When to Escalate

- Any report of a burning smell, scorch marks or melted terminals: treat as a safety case and follow the P1 process in the Ticket Escalation SOP.
- E1 or E6 that returns after a power-cycle and reseat.
- More than one T2 at the same address showing E4 at the same time after a firmware update.

8. Appendix A: Pre-Sale Compatibility Checklist

Customer Care uses this checklist before recommending the T2 to a customer who has not bought a thermostat yet. If any answer is "No" in the T2 column, recommend the T3 or a Certified Installer visit instead.

Question for the customer	T2 suitable if	If not
Is the current thermostat connected with thin, low-voltage wires?	Yes	Not compatible (line voltage).
Is there a wire on the C terminal, or can a PEK-1 be fitted?	Yes	Recommend the T3.
Does the system have more than 2 heat stages or 1 cool stage?	No	Recommend the T3.
Is it a heat pump with auxiliary or emergency heat?	No	Recommend the T3 with C-wire.
Does the customer want more than 2 room sensors?	No	Recommend the T3 (up to 6).
Does the home network offer 2.4 GHz?	Yes	Enable 2.4 GHz or use a Hub H1.

Table 8-1. T2 compatibility checklist.

9. Appendix B: Quick Answers for Agents

Can the T2 control a humidifier?

No. The T2 has no accessory terminals and no humidity sensor. Customers who need humidity control should be offered the T3.

Does the T2 work during a power outage?

No. The heating and cooling equipment does not run without power either. When power returns, the T2 restarts automatically with its schedule intact.

Can two T2 thermostats be used in the same home?

Yes, one per zone. Homes with two or more thermostats need a Hub H1 so that the app can show all zones in one home.

Can the T2 lock its controls?

Yes. Settings > Lock sets a 4-digit PIN. While locked, the control ring only shows the current temperature. The PIN can be removed from the app by the home Owner or an Admin.

Why does the display dim?

The backlight dims after 30 seconds without interaction to save power. Brightness and timeout can be changed under Settings > Display.

Revision History

Revision	Date	Change
3.1	March 2, 2026	Added PEK-1 free-of-charge window; firmware 4.8.2.
3.0	October 14, 2025	New error codes E8 and E9.
2.4	May 5, 2025	Clarified line-voltage warning.